applications beyond defense – into surveying, mapping, and environmental research. It also highlights early efforts toward autonomous flight through pre-programmed missions.

Chapter 4 marks the maturity of UAV technology, with drones gaining advanced surveillance, reconnaissance, and combat capabilities, grouping them into "Third-generation Drones." This chapter, emerging from the late 1990s to the early 2010s, describes drones such as the Predator and Global Hawk integrated with high-resolution sensors, secure data links, and long-endurance propulsion systems. The chapter discusses the evolution of unmanned combat aerial vehicles (UCAVs), emphasizing the convergence of defense strategy, automation, and ethical considerations in warfare.

Chapter 5 elaborates on fourth-generation drones. The fourth generation represents the shift from purely military use to widespread commercial adoption. With advancements in onboard processing, LiDAR, AI-assisted control, and real-time communication, drones began to perform complex civilian tasks such as precision agriculture, construction monitoring, disaster management, and logistics. This chapter explores the emergence of semi-autonomous flight, multi-sensor fusion, and improved energy efficiency. It also introduces global regulatory frameworks that enable the integration of drones into controlled airspace.

Chapter 6 explores the era of full autonomy, artificial intelligence, and swarm technology, coined as *"Fifth-generation Drones."* Fifth-generation drones are intelligent systems capable of independent decision-making, collaborative missions, and adaptive behavior. Key enabling technologies such as computer vision, edge computing, 5G communication, and blockchain-based security are discussed. The chapter also addresses futuristic applications in smart cities, autonomous logistics, defense